

Instantaneous contributions to the soft quark light-cone wave function: complete enumeration, diagrams and wave functions

Abstract

We enumerate every instantaneous contribution to the soft $q\bar{q}$ sector at $O(g^2)$ and give the corresponding diagram and light-cone wave function for each. There are two sources: the instantaneous gluon exchange, which is a single current–current operator whose cross terms generate five distinct topologies, and the instantaneous quark exchange, which generates three. A uniform vertex rule reduces all instantaneous-gluon graphs to one expression. Two of the topologies — the $\mathcal{J}_q \cdot \mathcal{J}_g$ cross terms, feeding $\mathcal{B}_3$ and $\mathcal{B}_1$ — do not appear anywhere in the current draft and were also absent from the corrected manuscript.

1 Where the instantaneous terms come from

Eliminating the constrained components A^- and ψ_- in $A^+ = 0$ gauge leaves two instantaneous operators.

(A) Instantaneous gluon exchange. This is a single current–current operator,

$$H_{\text{inst}}^{(A)} = \frac{g^2}{2} \int dx^- d^2x \mathcal{J}^a \frac{1}{(\partial^+)^2} \mathcal{J}^a, \quad \mathcal{J}^a = \underbrace{\rho^a}_{\text{valence}} + \underbrace{f^{abc} A_i^b \partial^+ A_i^c}_{\mathcal{J}_g^a} + \underbrace{\xi^\dagger t^a \xi}_{\mathcal{J}_q^a}. \quad (1)$$

Writing $\mathcal{J}$ as a sum makes the classification automatic: the cross terms are

cross term	content	status
$\rho \cdot \rho$	valence only	no soft dynamics
$\rho \cdot \mathcal{J}_g$	$H_{gg\text{-inst}}$ of the pure Yang–Mills paper	already known
$\rho \cdot \mathcal{J}_q$	Group A below	partly in the draft
$\mathcal{J}_q \cdot \mathcal{J}_q$	Group B below	missing from the draft
$\mathcal{J}_q \cdot \mathcal{J}_g$	Group C below	newly identified
$\mathcal{J}_g \cdot \mathcal{J}_g$	part of $H_{gggg} + H_{gggg\text{-inst}}$	pure glue

(B) Instantaneous quark exchange.

$$H_{ggq\bar{q}} = g^2 \int dx^- d^2x \xi^\dagger (\sigma^i A_a^i t^a) \frac{1}{2i\partial^+} (\sigma^j A_b^j t^b) \xi, \quad (2)$$

giving Group **D**.

2 Conventions

$\varepsilon_p = \mathbf{p}^2/2p^+$. The outgoing quark is (β, λ_1, l) and the outgoing antiquark (α, λ_2, m) , as in the ansatz. All the wave functions below are *first order* in perturbation theory — a single insertion of an instantaneous vertex — so in every case

$$\psi = \frac{\langle f | H_{\text{inst}} | i \rangle}{E_i - E_f}. \quad (3)$$

Diagram conventions.

$\longrightarrow$	quark (arrow = fermion flow)	$\sim$	gluon
$\equiv$	valence source ρ	$\vdash$	instantaneous exchange
$\vdots$	intermediate state		

The bar across a line is the standard light-cone marking for an instantaneous (non-propagating) exchange: it carries no energy denominator of its own, only the $1/(P^+)^2$ of Eq. (1) or the $1/2P^+$ of Eq. (2).

2.1 A uniform rule for the instantaneous gluon exchange

Every graph generated by (1) has the same form: two “charges” joined by a line carrying longitudinal momentum P^+ ,

$$\langle f | H_{\text{inst}}^{(A)} | i \rangle = \frac{g^2}{(2\pi)^3} \frac{\mathcal{Q}_{(1)}^a \mathcal{Q}_{(2)}^a}{(P^+)^2} \times \begin{cases} 1 & \text{if one charge is } \rho, \\ \delta^{(3)}(\Sigma_{\text{in}} - \Sigma_{\text{out}}) & \text{otherwise,} \end{cases} \quad (4)$$

with the charges

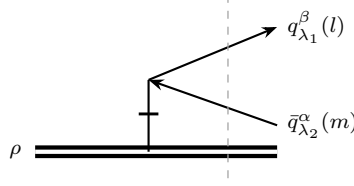
transition	$\mathcal{Q}^a$
valence	$\rho^a(\mp k)$
$q_\lambda^\beta(l) \rightarrow q_{\lambda'}^{\beta'}(l')$	$t_{\beta'\beta}^a \delta_{\lambda\lambda'}$
$\bar{q}_\lambda^\alpha(m) \rightarrow \bar{q}_{\lambda'}^{\alpha'}(m')$	$-t_{\alpha\alpha'}^a \delta_{\lambda\lambda'}$
$0 \rightarrow q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m)$	$t_{\beta\alpha}^a \delta_{\lambda_1\lambda_2}$
$g_i^b(p) \rightarrow g_j^c(q)$	$\frac{if^{abc}(p^+ + q^+) \delta_{ij}}{2\sqrt{p^+q^+}}$
$0 \rightarrow g_i^b(p) g_j^c(q)$	$\frac{if^{abc}(q^+ - p^+) \delta_{ij}}{2\sqrt{p^+q^+}}$

The two last entries are read off from Eqs. (D.4) and (D.5) of the pure Yang–Mills paper; the quark entries are fixed by Eq. (22) of the notes. Note that $8\pi^3 = (2\pi)^3$, so the two calibrations already agree.

One caveat on normalization. Rule (4) is calibrated on the two cases in which one charge is the valence ρ — Eq. (22) of the notes and Eqs. (D.4)/(D.5) of the pure gauge paper — and those two agree with each other. For Groups **B** and **C**, where *both* charges are soft, the relative factor between the “ ρ present” and “ ρ absent” cases in (4) follows from writing (1) in momentum space as $\frac{g^2}{2} \int \frac{d^3P}{(2\pi)^3} (P^+)^{-2} \mathcal{J}^a(-P) \mathcal{J}^a(P)$, but I have not been able to check it against an independently normalized expression in either draft. The colour factors, P^+ assignments, relative signs and energy denominators below are unaffected; *confirm the overall constant of Groups B and C against your own derivation of $H_{\text{inst}}^{(A)}$ before using them.*

3 Group A: valence $\leftrightarrow$ soft quark ($\rho \cdot \mathcal{J}_q$)

A1. $|0\rangle \rightarrow |q\bar{q}\rangle$

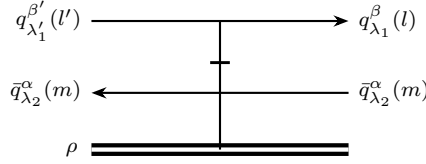


$P^+ = k^+ = l^+ + m^+$, $E_i = 0$, $E_f = \varepsilon_l + \varepsilon_m$:

$$\psi_{A1} = -\frac{g^2 t_{\beta\alpha}^a \rho^a(-k) \delta_{\lambda_1\lambda_2}}{(2\pi)^3 (k^+)^2 (\varepsilon_l + \varepsilon_m)}, \quad k = l + m. \quad (5)$$

This is Eq. (22) of the notes. It feeds $\mathcal{A}_1$, and in the combination with the non-instantaneous graph it cancels the $\delta_{\lambda_1\lambda_2}/(k^+)^2 \varepsilon_{q\bar{q}}$ piece exactly, leaving the compact two-term $\mathcal{A}_1$.

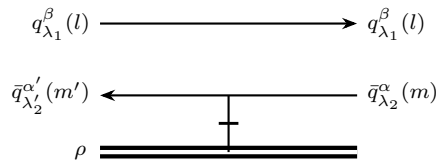
A2. $|q\bar{q}\rangle \rightarrow |q\bar{q}\rangle$, quark leg scatters off the valence



$P^+ = l'^+ - l^+$, $E_i - E_f = \varepsilon_{l'} - \varepsilon_l$ (the antiquark is a spectator):

$$\psi_{A2} = \frac{g^2 t_{\beta\beta'}^a \rho^a(l' - l) \delta_{\lambda_1\lambda_1'} \delta^{\alpha\alpha'} \delta_{\lambda_2\lambda_2'} \delta^{(3)}(m - m')}{(2\pi)^3 (l'^+ - l^+)^2 (\varepsilon_{l'} - \varepsilon_l)}. \quad (6)$$

A3. $|q\bar{q}\rangle \rightarrow |q\bar{q}\rangle$, antiquark leg scatters off the valence

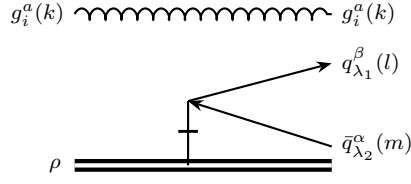


The antiquark charge carries the extra minus sign of the rule:

$$\psi_{A3} = -\frac{g^2 t_{\alpha'\alpha}^a \rho^a(m' - m) \delta_{\lambda_2\lambda_2'} \delta^{\beta\beta'} \delta_{\lambda_1\lambda_1'} \delta^{(3)}(l - l')}{(2\pi)^3 (m'^+ - m^+)^2 (\varepsilon_{m'} - \varepsilon_m)}. \quad (7)$$

A2 and A3 are the eikonal colour rotation of a soft quark in the valence background. They populate the $b^\dagger b$ and $d^\dagger d$ structures, which are *absent from the ansatz*; within the $q\bar{q}$ sector they contribute to $\mathcal{D}_3$ through the spectator delta.

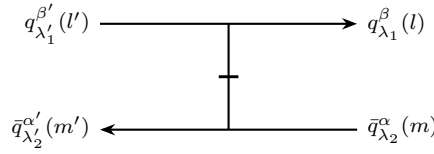
A4. $|g\rangle \rightarrow |g q \bar{q}\rangle$ (spectator gluon)



Identical to A1 times the spectator delta. It is *disconnected* and is already accounted for by the $\mathcal{A}_1$ term in the action of Ω on $|g\rangle$; it must be subtracted, not added, when reading off $\mathcal{B}_3$.

4 Group B: soft quark $\leftrightarrow$ soft quark ($\mathcal{J}_q \cdot \mathcal{J}_q$)

B1. $|q\bar{q}\rangle \rightarrow |q\bar{q}\rangle$, instantaneous gluon between the two legs

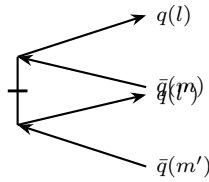


$P^+ = l'^+ - l^+ = m^+ - m'^+$; the product of a quark charge and an antiquark charge carries one minus sign:

$$\psi_{B1} = - \frac{g^2 t_{\beta\beta'}^a t_{\alpha'\alpha}^a \delta_{\lambda_1\lambda_1'} \delta_{\lambda_2\lambda_2'} \delta^{(3)}(l + m - l' - m')}{(2\pi)^3 (l'^+ - l^+)^2 (\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_l - \varepsilon_m)}. \quad (8)$$

This is the instantaneous $\mathcal{J}_q \cdot \mathcal{J}_q$ exchange flagged as missing from the $\mathcal{D}_3$ sector. Together with the t -channel one-gluon exchange it must satisfy $X(m', l', m, l) + X^\dagger(m, l, m', l') = 0$, since the s -channel graph already saturates the unitarity relation for $\mathcal{D}_3$.

B2. $|0\rangle \rightarrow |q\bar{q} q\bar{q}\rangle$

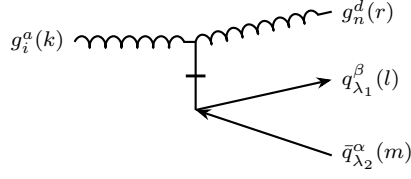


Both currents create a pair. This is a legitimate $O(g^2)$ transition, but the structure $b^\dagger d^\dagger b^\dagger d^\dagger$ with no gluon operator is *not in the ansatz*; it first matters when Ω acts on states containing two pairs. Recorded here for completeness.

5 Group C: soft quark $\leftrightarrow$ soft gluon ($\mathcal{J}_q \cdot \mathcal{J}_g$)

These two topologies appear nowhere in the draft, and I did not include them in the corrected manuscript either. They arise because $\mathcal{J}_g$ — the soft gluon colour charge density in Eq. (1) — can be contracted against the soft quark current, not only against ρ . They are the same order as every other instantaneous term and feed $\mathcal{B}_3$ and $\mathcal{B}_1$.

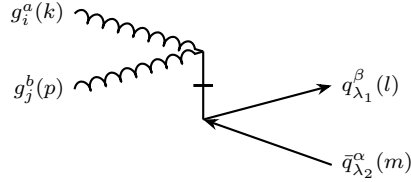
C1. $|g\rangle \rightarrow |q\bar{q}g\rangle$ (feeds $\mathcal{B}_3$)



Charges: pair creation $t_{\beta\alpha}^b \delta_{\lambda_1\lambda_2}$ and gluon transition $if^{bad}(k^+ + r^+) \delta_{in}/2\sqrt{k^+r^+}$; $P^+ = l^+ + m^+ = k^+ - r^+$; $E_i - E_f = \varepsilon_k - \varepsilon_l - \varepsilon_m - \varepsilon_r$:

$$\psi_{C1} = \frac{i g^2 f^{bad} t_{\beta\alpha}^b \delta_{\lambda_1\lambda_2} \delta_{in} (k^+ + r^+) \delta^{(3)}(k - l - m - r)}{2 (2\pi)^3 \sqrt{k^+ r^+} (l^+ + m^+)^2 (\varepsilon_k - \varepsilon_l - \varepsilon_m - \varepsilon_r)}. \quad (9)$$

C2. $|gg\rangle \rightarrow |q\bar{q}\rangle$ (feeds $\mathcal{B}_1$)



Charges: pair creation $t_{\beta\alpha}^c \delta_{\lambda_1\lambda_2}$ and two-gluon annihilation, the conjugate of the creation entry, $-if^{cab}(p^+ - k^+) \delta_{ij}/2\sqrt{k^+p^+}$; $P^+ = k^+ + p^+$:

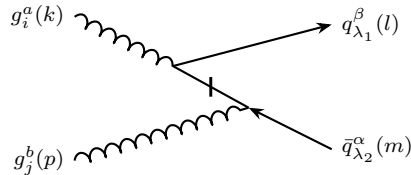
$$\psi_{C2} = - \frac{i g^2 f^{cab} t_{\beta\alpha}^c \delta_{\lambda_1\lambda_2} \delta_{ij} (p^+ - k^+) \delta^{(3)}(k + p - l - m)}{2 (2\pi)^3 \sqrt{k^+ p^+} (k^+ + p^+)^2 (\varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m)}. \quad (10)$$

The remaining possibility, $|0\rangle \rightarrow |q\bar{q}gg\rangle$, is forbidden: with no valence charge to absorb momentum, four outgoing quanta with $p^+ > 0$ cannot conserve p^+ .

6 Group D: instantaneous quark exchange ($H_{ggq\bar{q}}$)

Here the exchanged line is a fermion. The rule is the standard light-cone one: the instantaneous fermion carries $\gamma^+/2P_{\text{inst}}^+$, which in the two-component form becomes $\sigma^i (2P_{\text{inst}}^+)^{-1} \sigma^j$ between the two gluon attachments, with P_{inst}^+ the $+$ momentum flowing along the barred line.

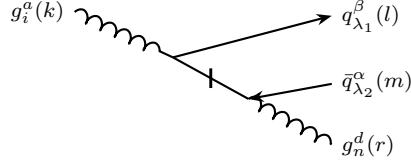
D1. $|gg\rangle \rightarrow |q\bar{q}\rangle$ (feeds $\mathcal{B}_1$)



The two orderings of the gluon attachments give $P_{\text{inst}}^+ = l^+ - k^+$ and $l^+ - p^+$:

$$\psi_{D1} = \frac{g^2 \delta^{(3)}(k + p - l - m)}{16\pi^3 \sqrt{k^+ p^+} (\varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m)} \chi_{\lambda_1}^\dagger \left[\frac{(t^a t^b)_{\beta\alpha} \sigma^i \sigma^j}{l^+ - k^+} + \frac{(t^b t^a)_{\beta\alpha} \sigma^j \sigma^i}{l^+ - p^+} \right] \chi_{\lambda_2}. \quad (11)$$

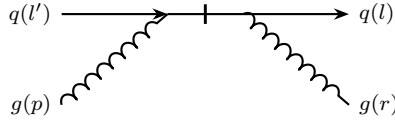
D2. $|g\rangle \rightarrow |q\bar{q}g\rangle$ (feeds $\mathcal{B}_3$)



Now one gluon is incoming and one outgoing, so $P_{\text{inst}}^+ = l^+ + r^+$ when the outgoing gluon attaches on the quark side and $-(m^+ + r^+)$ when it attaches on the antiquark side:

$$\psi_{\text{D2}} = \frac{g^2 \delta^{(3)}(k - l - m - r)}{16\pi^3 \sqrt{k^+ r^+} (\varepsilon_k - \varepsilon_l - \varepsilon_m - \varepsilon_r)} \chi_{\lambda_1}^\dagger \left[\frac{(t^d t^a)_{\beta\alpha} \sigma^n \sigma^i}{l^+ + r^+} - \frac{(t^a t^d)_{\beta\alpha} \sigma^i \sigma^n}{m^+ + r^+} \right] \chi_{\lambda_2}. \quad (12)$$

D3. $|qg\rangle \rightarrow |qg\rangle$



Instantaneous quark exchange in the qg channel. The structure is $a^\dagger a b^\dagger b$, which is *not in the ansatz*: it contributes only when Ω acts on states containing both a soft gluon and a soft quark. Recorded for completeness.

7 Summary

	transition	origin	feeds	in the draft?
A1	$ 0\rangle \rightarrow q\bar{q}$	$\rho \cdot \mathcal{J}_q$	$\mathcal{A}_1$	yes, Eq. (22)
A2	$q\bar{q} \rightarrow q\bar{q}$ (quark leg)	$\rho \cdot \mathcal{J}_q$	$b^\dagger b, \mathcal{D}_3$	no
A3	$q\bar{q} \rightarrow q\bar{q}$ (antiquark leg)	$\rho \cdot \mathcal{J}_q$	$d^\dagger d, \mathcal{D}_3$	no
A4	$g \rightarrow g q\bar{q}$ (spectator)	$\rho \cdot \mathcal{J}_q$	disconnected ($\mathcal{A}_1$)	implicitly
B1	$q\bar{q} \rightarrow q\bar{q}$	$\mathcal{J}_q \cdot \mathcal{J}_q$	$\mathcal{D}_3$	no
B2	$ 0\rangle \rightarrow q\bar{q} q\bar{q}$	$\mathcal{J}_q \cdot \mathcal{J}_q$	beyond the ansatz	no
C1	$g \rightarrow q\bar{q} g$	$\mathcal{J}_q \cdot \mathcal{J}_g$	$\mathcal{B}_3$	no — new
C2	$gg \rightarrow q\bar{q}$	$\mathcal{J}_q \cdot \mathcal{J}_g$	$\mathcal{B}_1$	no — new
D1	$gg \rightarrow q\bar{q}$	$H_{ggq\bar{q}}$	$\mathcal{B}_1$	no
D2	$g \rightarrow q\bar{q} g$	$H_{ggq\bar{q}}$	$\mathcal{B}_3$	no
D3	$qg \rightarrow qg$	$H_{ggq\bar{q}}$	beyond the ansatz	no

Of the eleven instantaneous topologies, exactly one — A1 — appears in the current draft. Four (A4, B1, D1, D2) were restored in the corrected manuscript. Two (C1, C2) are identified here for the first time and need to be added to $\mathcal{B}_3$ and $\mathcal{B}_1$. Three (A2, A3, D3) populate operator structures that the ansatz does not contain, and B2 likewise; they are inert for the matching sectors used so far, but must be restored before Ω is applied to states containing both soft gluons and soft quarks.

Two structural remarks. First, writing the instantaneous gluon exchange as the single current-current operator (1) rather than as a list of separate Hamiltonians is what makes the enumeration exhaustive: one simply expands $\mathcal{J} = \rho + \mathcal{J}_g + \mathcal{J}_q$ and reads off the cross terms. The two missing topologies C1 and C2 are exactly the ones that a list-of-Hamiltonians organization loses. Second,

every graph in Groups A–C collapses onto the single rule (4), so only the charges and P^+ need to be identified in each case.